

Quy-Sang Mai, CFA

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(+84) 901.350.009 - Ho Chi Minh City, Vietnam

Research interests: ML-driven signal combination for abnormal returns in emerging equity markets; investor sentiment and earnings-surprise prediction.

EDUCATION

MSc, Applied Mathematics (Quantitative & Computational Finance) 2016 - 2019

University of Science, VNU-HCM - Capstone graded 10/10; Stochastic Calculus 9.5/10, Probability & Statistics 9.0/10

Bachelor of Business (Economics and Finance) 2012 - 2015

RMIT University - Awarded with Distinction

Mathematics (specialised), High School for the Gifted, VNU-HCM 2009 - 2012

RESEARCH EXPERIENCE

Capstone Project (graded 10/10) - University of Science, VNU-HCM Jan - Mar 2018

Built a three-layer system for VN30 equities - gradient-boosted decision-tree 'experts' over ~40 technical signals, combined by an online expert-weighting layer ([Creamer & Freund 2010](#)) with a risk overlay - to predict the sign of next-day abnormal returns vs the index. Out-of-sample (2015-17) it beat the VN30 on risk-adjusted measures (Information / Sharpe ratios ~2-4), with a transaction-cost sensitivity analysis confirming outperformance persisted to ~2 bps per trade. The same signal-combination approach is what the proposed PhD would extend to text- and LLM-derived sentiment signals for earnings-surprise prediction. Supervised by [Dr Minh Dang](#) (State Street, Hong Kong).

Master's Thesis - University of Science, VNU-HCM 2019

"Smile-adjusted delta hedging for options improved with boosting." Added a gradient-boosted-regression-tree step to the minimum-variance delta of [Hull and White \(2017\)](#) ; on ~1.3M S&P 500 index option records (OptionMetrics, 2015-17) it delivered a statistically significant out-of-sample reduction in hedging error (Newey-West t-tests) for put options. Supervised by [Dr Minh Dang](#) .

Research Assistant - RMIT University Vietnam Aug 2016 - Mar 2017

Co-authored learning-and-teaching research with [A/Prof. Mathews Nkhoma](#) (now Associate Deputy Vice-Chancellor, RMIT University): literature review and qualitative & quantitative data analysis, resulting in the peer-reviewed conference paper below.

PUBLICATIONS

Nkhoma, C. A., Nkhoma, M., Ulhaq, I., & Mai, S. Q. (2017). [Enhancing students' learning through early class preparation](#). Proceedings of InSITE 2017, 85-95. [doi:10.28945/3756](#)

TEACHING

Teaching Assistant - "Derivatives Pricing in Practice" (graduate course) 2017

University of Science, VNU-HCM. Supported delivery of the graduate derivatives-pricing course under [Dr Minh Dang](#) .

PROFESSIONAL EXPERIENCE

Ho Chi Minh City Securities Corporation (HSC) 2017 - Present

One of Vietnam's leading brokerage and investment-banking firms.

Manager / Senior Manager - Structured Products & Quantitative Trading Jan 2024 - Present

- Lead quantitative research - developing and backtesting models to construct equity portfolios optimised for absolute and risk-adjusted returns;
- Oversee the structured-products (covered warrants) desk - pricing, market-making, hedging and reporting;
- Mandated to build out and lead structured-products trading (equity options and other new structured products) as they launch in the Vietnamese market.

Senior / Covered Warrant Trader; ETF Trader Sep 2017 - Dec 2023

- Product development, pricing, market-making and reporting for HSC's covered-warrant and ETF portfolios (authorised participant); backup coverage for index arbitrage.

Investment Analyst - Saigon Asset Management (SAM) Aug 2014 - Aug 2015

Research and valuation of Vietnamese listed companies (DCF and comparables) for funds with peak AUM over USD\$200m.

TECHNICAL SKILLS

Derivatives & options (pricing, Greeks, hedging) | Econometrics, abnormal-return analysis & time-series | Supervised ML (gradient boosting, online learning) | Portfolio construction, out-of-sample & after-cost backtesting | Python (pandas, scikit-learn)

CREDENTIALS

CFA Charterholder (2022) - Fund Management Professional License, Vietnam (2023)

REFEREES

Available on request.